GlycoRenal Twin

Clinical report 17 Oct 2026 18:35

Patient IND-0042 | 54 y M | T2DM 9 years | Doctor: Dr. Anjali Mehta

Predicted risk

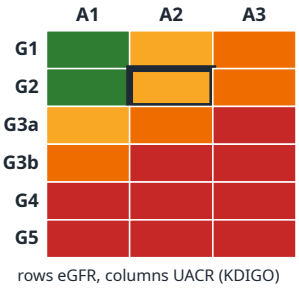
3-year risk of kidney disease worsening within 3 years: 42% (high).

Raising the risk: eGFR, Albuminuria (UACR), Kidney polygenic score (simulated).

Model: dkd_progression@2026-10-16

HbA1c forecast in 3 months: 8.3% (80% range 7.9 to 8.7%).

KDIGO G2-A2 (moderate risk). eGFR slope -8.02 per year; change over 12 months -11%.



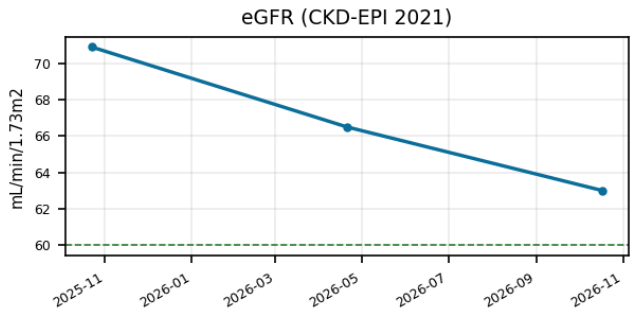
rows eGFR, columns UACR (KDIGO)

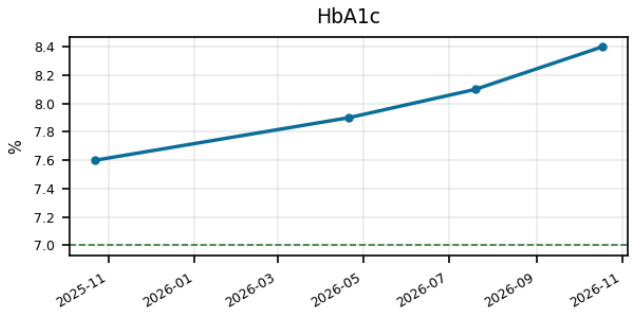
Latest results

Test	Value	Date	Age	Status
HbA1c	8.4 %	17 Oct 2026	0 days	current
Serum creatinine	1.34 mg/dL	17 Oct 2026	0 days	current
eGFR (CKD-EPI 2021)	63 mL/min/1.73m2	17 Oct 2026	0 days	current
Urine albumin/creatinine ratio	85 mg/g	17 Oct 2026	0 days	current
Potassium	4.6 mmol/L	17 Oct 2026	0 days	current
LDL cholesterol	118 mg/dL	17 Oct 2026	0 days	current
Blood pressure	146/88 mmHg	target < 130/80		above target
BMI / weight	27.9 kg/m2, 79.8 kg			
Medicine adherence	66.7% (7 d), 58.9% (30 d)			low

Genetic markers (simulated): Kidney polygenic risk score: 82nd percentile: higher inherited kidney risk; TCF7L2 rs7903146 (T2D risk gene): 2 risk alleles; CYP2C9 metaboliser: intermediate metaboliser; SLC22A1 (OCT1) reduced-function alleles: 0 risk alleles.

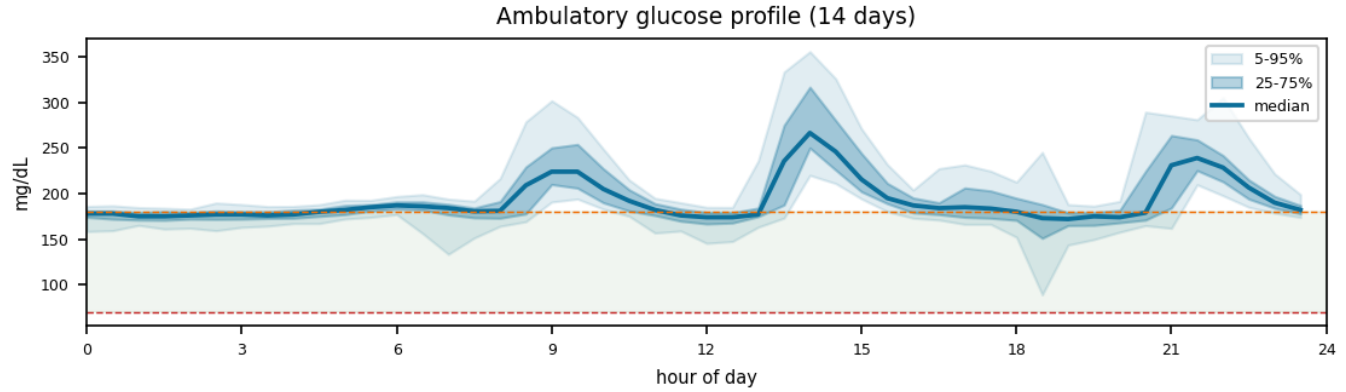
Trends





CGM summary (14 days)

Time in range 41.6% (target > 70%); below 70 mg/dL 0.1% (target < 4%); below 54 0.0%; above 180 58.3%; above 250 7.5%. Mean 193.7 mg/dL, GMI 7.94%, CV 17.3%, data sufficiency 99.5%.



Personal twin (knows this patient: 92%)

Glucose twin tuned on 13.8 days of CGM and 48 meals: +12.9 mg/dL per 10 g carbs, peak 50 min after eating, 15-min walk lowers by 15 mg/dL; 2 katori rice: +141 mg/dL. Replays the last 3 days within 8.2 mg/dL on average.
Own eGFR slope -3.57 mL/min/yr (80% range -6.33 to -0.8; population -2.06), from 3 results. eGFR < 60: 1.4 y; < 45: 5.6 y; < 30: 9.8 y.

Treatment options, ranked by predicted reduction in 3-year kidney risk

#	Option	3-yr risk	Evidence
1	Add GLP-1 receptor agonist (e.g. semaglutide)	42% to 30%	FLOW, Perkovic NEJM 2024 (verify)
2	Bring BP to target (< 130/80)	42% to 33%	KDIGO 2021 BP in CKD guideline (verify)
3	Add finerenone 10-20 mg	42% to 34%	FIDELIO-DKD, Bakris NEJM 2020 (verify)
4	Lose 5% of body weight	42% to 40%	Look AHEAD, Wing Diabetes Care 2011 (verify)
5	Add sulfonylurea (e.g. gliclazide)	42% to 41%	ADA Standards of Care 2026 sec 9 (verify)
6	Start basal insulin	42% to 42%	ADA Standards of Care 2026 sec 9 (verify)
7	Add DPP-4 inhibitor (e.g. linagliptin)	42% to 42%	ADA Standards of Care 2026 sec 9 (verify)
8	Walk 2,000 more steps a day	42% to 42%	Umpierre JAMA 2011 (verify)

Combined plan (Add GLP-1 receptor agonist (e.g. semaglutide), Add finerenone 10-20 mg, Bring BP to target (< 130/80)): 42% to 20%.

Not eligible or already in place: Add SGLT2 inhibitor (e.g. dapagliflozin 10 mg) (Already on an SGLT2 inhibitor (continue until dialysis/transplant)); Start / maximise ACE inhibitor or ARB (e.g. telmisartan) (Already on ACEi or ARB: titrate to max tolerated dose; never combine ACEi with ARB); Add metformin (Already on metformin); Stop smoking (Non-smoker)

Current medicines

Metformin 1000 mg at 08:00, 20:00; Telmisartan 40 mg at 08:00; Dapagliflozin 10 mg at 08:00

Active alerts

[critical] Glucose 52 mg/dL (below 54): severe hypoglycaemia
[warning] Glucose 64 mg/dL (below 70): hypoglycaemia
[warning] Low glucose likely within 30 min (now 126, forecast 127, trend 78 mg/dL)
[warning] Albuminuria rose from A1 to A2 (UACR 85 mg/g)
[warning] Glucose 251 mg/dL (above 250)
[warning] Glucose spike above 250 likely within 90 min (probability 64%, now 153 mg/dL; meal of 99 g carbs in the last 2 h)
[warning] Medicine adherence 67% in the last 7 days

Clinical summary (AI: primary (azure_ai/kimi-k2.6))

Patient IND-0042 is a 54-year-old male with type 2 diabetes for 9 years, currently at KDIGO G2-A2 with moderate risk (eGFR 63.0 mL/min/1.73m^2, UACR 85 mg/g). Glycaemic control is suboptimal (HbA1c 8.4%, target <7.0%) and blood pressure is above goal (146/88 mmHg vs target <130/80). The eGFR trend shows decline of -8.02 per year. The twin predicts a 42% risk of kidney disease worsening within 3 years (high risk band), driven by eGFR, albuminuria, and kidney polygenic score. Top ranked interventions are: add GLP-1 receptor agonist (semaglutide) reducing risk to 30% and HbA1c to 7.4% (FLOW); achieve BP target <130/80 reducing risk to 33% (KDIGO 2021); and add finerenone 10-20 mg reducing risk to 34% (FIDELIO-DKD).